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A Demo of Cache-Augmented Generation (CAG) in an LLM

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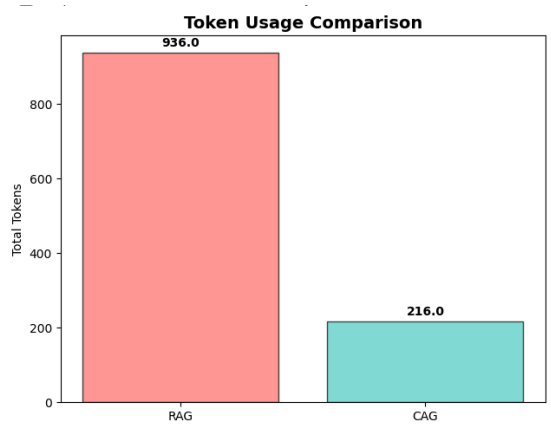
A Demo of Cache-Augmented Generation (CAG) Using an Open-Source LLM



This is a simple demo of Cache-Augmented Generation (CAG) using an Open-Source LLM.

CAG preloads relevant knowledge into a language model's context, allowing for faster and more efficient question-answering without real-time document retrieval. This project demonstrates a simple implementation of CAG and its 76% token reduction usage compared to RAG.

Article: https://medium.com/@ronantech/cache-augmented-generation-cag-in-llms-a-step-by-step-tutorial-6ac35d415eec

Google Colab: https://colab.research.google.com/drive/1-0eKIu6cGAZ47ROKQaF6EU-mHtvJBILV?usp=sharing

This project was described as a reliable example of CAG in 2 Reddit threads:
https://www.reddit.com/r/LangChain/comments/1jh759u/cache_augmented_generation/
https://www.reddit.com/r/Rag/comments/1i5l2ot/dont_do_rag_its_time_for_cag/

Overview

The notebook `cag_demo.ipynb` showcases the core steps of CAG:

1. Loading the Mistral model and tokenizer
2. Preloading that knowledge into the model's context using a `DynamicCache`
3. Answering user queries by referencing the cached knowledge, without real-time retrieval
4. Saving the context cache on disk, then reloading it for another chat session

Setup

Prerequisites

- Python 3.7+
- PyTorch 1.13+
- Transformers 4.28+
- Hugging Face account with access to the `mistralai/Mistral-7B-Instruct-v0.1` model

Installation

1. Clone this repository:

```
git clone https://github.com/yourusername/cache-augmented-generation.git
cd cache-augmented-generation
```

2. Install the required packages:

```
pip install torch transformers
```

Usage

1. Open the `cagdemo.ipynb` notebook in Jupyter, VS Code, or Google Colab.

Releases

No releases published

README

Packages

No packages published

Languages

- **Jupyter Notebook** 100.0%